NETWORKING

What Is Networking

It's a interconnected computing device that can exchange data and share the resources within a network.

There Are Two Types of Addresses

1. physical address (mac address – 48 bit)
2. logical address (ipv4 – 32bit / ipv6 – 128bit)

We Are Working On The Ipv4 Address – 32 Bit

8 bit	8 bit	8 bit	8 bit
11111111	11111111	11111111	11111111
255	255	255	255

1	1	1	1	1	1	1	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

1.we can denote host by --------> 0
2.we can denote network by ---> 1

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what is network address:

The first address of the ip address

what is broadcast address:

The last address of the ip address

CLASSES

1. 0------126 (N8)--------> N H H H ---------> 256*256*256 ------> 16,777,216
2. 128----191 (N16)-----> N N H H ---------> 256*256 ----------> 65,536
3. 192----223 (N24)-----> N N N H ---------> 256 ----------> 256
4. 224----239 ----------> special purpose
5. 240----255----------> future used

{ Note:- 127 is the loopback ip / systems ip }

CIDR (Classless Inter-Domain Routing)

1. CIDR (Classless Inter-Domain Routing) is a method used to divide IP addresses into flexible network sizes.
2. It helps in saving IP addresses and makes routing on the internet more efficient.

How To Calculate Hosts

syntax: no of useful hosts = 2^(host bit) – 2

Note:

1. host bit ----> 32-prefix length
2. -2 ----------> one for network and another for broadcast address
3. 1 - denoted by network
4. 0 - denoted by hosts

1.Class A: N(8) ----------> N H H H /8

subnet mask ----> 255 0 0 0
no of hosts ------> 2^n -2
2^24 -2
16777721 hosts

2.Class B: N(16) ---------> N N H H /16

subnet mask ----> 255 255 0 0
no of hosts ------> 2^n-2
2^16-2
65534 hosts

3.Class C: N(24) ---------> N N N H /24

subnet mask ----> 255 255 255 0
no of hosts ------> 2^n-2
2^8-2
254 hosts

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CIDR Examples:

12.0.0.0/26

11111111.11111111.1111111.11000000

2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0
128 64 32 16 8 4 2 1

1. subnet mask ----> 255.255.255.192
2. subnet id ------> 12.0.0.0/26
3. no of hosts ----> 2^n-2 = 2^6-2 = 64-2 = 62
4. networks ------> 2^n = 2^2 = 4

subnets:

1. 12.0.0.0/26
2. 12.0.0.64/26
3. 12.0.0.128/26
4. 12.0.0.192/26 -------> last subnet

Broadcast Addresses:

1. 12.0.0.63/26
2. 12.0.0.127/26
3. 12.0.0.191/26
4. 12.0.0.255/26

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Note:

1. Understanding CIDR (/26)

/26 means:

- Total bits = 32
- Network bits = 26
- Host bits = 6

2. Binary of Subnet Mask

11111111.11111111.11111111.11000000

1. Convert last octet: 128 + 64 = 192
2. Subnet Mask = 255.255.255.192

3.Find Block Size (calculate hosts)

Number of useful hosts = $2^n - 2$

$= 2^6 - 2$

$= 64 - 2$

$= 62$ hosts

Subnets will increase by 64 in subnet range

4.Number of Subnets

1. Default Class A = /8
   Given = /26
2. Borrowed bits = 26 − 24 = 2 bits
3. Subnets = $2^2 = 4$

5. What is Subnet mask

Subnet mask tells us which part of an IP address is:

- Network part
- Host part

Practice Example

1. 10.0.0.0/20
2. 192.168.0.0/25
3. 12.0.0.0/18
4. 172.168.0.0/21

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Private IP Ranges

1. Class A --------- 10.0.0.0 -------- 10.255.255.255 (for big organizations)
2. Class B --------- 172.16.0.0 ----- 172.31.255.255 (Medium size companies)
3. Class C --------- 192.168.0.0 ---- 192.168.255.255 (Home / WiFi / small office)

DIFFERENCE BETWEEN THE PUBLIC IP AND THE PRIVATE IP

Public IP

1. Assigned by Internet Service Provider (ISP).
2. Globally unique (no two devices on the internet can have the same public IP).
3. Routable on the internet (can communicate directly over the internet).
4. Often comes with extra cost (paid service).

Private IP

1. Assigned by system admin / router / DHCP inside the LAN.
2. Locally unique only (can be repeated in different networks).
3. Not routable on the internet directly (requires NAT to communicate outside).
4. Free of cost (anyone can use private IP ranges).

OSI MODEL (open system Interconnection)

OSI model is a open system Interconnection a 7-layer model that the computer systems used to communicate over the network.

7. Application layer--------> Provides network services to users/app (http/https)
6. Presentation layer ------> Translates, encrypts, compresses data
5. session layer ----------> Manages communication sessions
4. transport layer ----------> Provides reliable data transfer (TCP/UDP)
3. network layer ----------> Handles routing and IP addressing (IPs)
2. data link layer ----------> Transfers data using MAC addresses
1. physical layer ----------> Sends raw bits through cables/signals

WORKING OF 7 LAYERS

7. Application Layer

- Closest layer to the user
- Used by applications like browsers, email, WhatsApp

Examples:

- HTTP
- HTTPS
- FTP
- DNS

6. Presentation Layer

- Converts data format
- Encrypts and decrypts data
- Compresses files

Example:

- SSL/TLS encryption

5. Session Layer

- Creates and manages sessions between devices

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- Keeps connection active

Example:

- Login session in websites

4. Transport Layer

- Ensures reliable communication
- Breaks data into segments
- Error checking

Protocols:

- TCP
- UDP

3. Network Layer

- Finds best path for data
- Uses IP addresses

Devices:

- Routers

Protocol:

- IP

2. Data Link Layer

- Uses MAC addresses
- Detects errors in frames

Devices:

- Switches

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1. Physical Layer

- Actual transmission of bits
- Uses cables, radio signals, fiber

Devices:

- Hub
- Cable
- Repeater

When sending data:

Application → Presentation → Session → Transport → Network → Data Link → Physical

When receiving data:
Physical → Data Link → Network → Transport → Session → Presentation → Application

Example:
Sending a message on WhatsApp:

1. Application → WhatsApp app creates message
2. Presentation → Encrypts message
3. Session → Maintains chat session
4. Transport → Splits message into packets
5. Network → Adds IP address
6. Data Link → Adds MAC address
7. Physical → Sends through Wi-Fi/mobile network

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